

DONFACK FORTUNE

Renewable Energy Specialist · Climate Researcher · Public Communicator

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Mechanical engineer and renewable-energy researcher with field-validated work in off-grid energy systems, climate-resilient agricultural technology and applied climate communication. Founder of **The REM** — a newsletter reaching 3,200+ subscribers — translating technical energy research into policy-relevant insight for West and Central Africa.

EXPERIENCE

Renewable Energy Researcher & Communicator

2023 – Present

The REM Newsletter & Medium — Self-initiated, Douala

- Founded and edit **The Renewable Energy Mall (REM)**, a newsletter reaching 3,200+ engineering and policy subscribers, converting applied climate research into accessible communication aligned with SDG 7 & 13.
- Authored research-grade articles on Cameroon's energy transition, NDC targets (32% GHG reduction by 2030) and off-grid solar deployment barriers — cited by climate practitioners across West and Central Africa.
- Analysed national energy data (78% biomass dependence; 0.01% installed solar capacity, 2018) to propose evidence-based rural electrification pathways.

Undergraduate Research Lead, Renewable Energy Systems

Jan – Mar 2022

University of Douala — Electrical Department

- Directed design, prototyping and physical deployment of the **Pedal-Power Mobile Phone Charger**, a zero-emission off-grid solution that reduced the campus energy shortfall by 25%.
- Conducted system-level performance testing with Arduino and Simulink/MATLAB — a 20% gain in electrical efficiency over baseline benchmarks.
- Documented methodology, failure modes (FMEA) and scalability, positioning findings for Cameroon's off-grid rural electrification planning under the 2021 NDC framework.

Mechanical Design Lead, Agricultural Climate Adaptation

Mar – May 2022

Community Project — Maroua, Far North Region

- Designed and built a high-precision **Beans Unwrapping Machine** for smallholder farmers, raising post-harvest processing efficiency by 25% under climate-stress conditions (up to 45 °C).
- Applied ANSYS vibration simulation during fabrication to ensure structural resilience, cutting equipment breakdown frequency by 30% and lowering maintenance costs for low-income households.

Low-Carbon Infrastructure Engineer

Apr – Aug 2021

Cameroon Railway Company (CAMRAIL) — Braking System Dept.

- Led FEA and CFD optimisation of the Rk7 sole holder and braking triangle on intercity trains — a 35% braking-performance gain supporting Cameroon's national low-carbon rail mandate.
- Produced full technical drawing sets (AutoCAD, SolidWorks) and presented deliverables to the CAMRAIL Tech Office.

EDUCATION

Master's in Mechanical Engineering

ENSET, Douala · 2022 – 2024 · GPA 3.52 / 5.0 · Honours

Bachelor's in Mechanical Engineering — Mechanical Construction

ENSET, Douala · 2019 – 2022 · GPA 3.63 / 5.0 · Honours

RESEARCH FOCUS

SDG 7 — Affordable & Clean Energy. Off-grid rural electrification and solar-deployment barriers in Sub-Saharan Africa.

SDG 13 — Climate Action. GHG-reduction strategy, Cameroon NDC targets, climate-resilient infrastructure design.

SDG 2 — Zero Hunger. Climate-adaptive agricultural technology and food-security engineering for the Far North.

NDC Cameroon. 32% GHG reduction by 2030; biomass-to-clean-energy transition analysis.

SKILLS

Climate & Energy

- Climate risk assessment
- GHG reduction strategy
- Renewable energy design
- Off-grid systems
- NDC analysis
- Food-security eng.

Research & Communication

- Applied research
- Climate writing
- Data visualisation
- Stakeholder reporting
- Newsletter editing

Engineering Tools

- ANSYS
- MATLAB / Simulink
- SolidWorks
- Arduino
- FEA / CFD
- Python
- Excel

Languages

- French (native)
- English (professional)

CERTIFICATIONS

- Failure Mode & Effects Professional — AIGPE 2024
- MATLAB Professional — MathWorks 2024

3,200+

newsletter subscribers · applied climate & energy articles